

Abdelilah Younsi

AI Engineer/ Data Scientist

📍 Strasbourg ✉ abdelilah.younsi@polytechnique.edu ☎ +33775840612 🌐 AbdelilahYounsi 🌐 Website

Skills

Software Engineering: Python, FastAPI, Flask

ML/DL Libraries: Pandas, scikit-learn, XGBoost, Optuna, PyTorch, PyTorch Lightning, Accelerate

MLOps & DevOps: Git, Docker, MLflow, Kubeflow, Airflow, Azure ML, Azure DevOps, Azure pipelines

Cloud & Data: Azure, AWS, Databricks, Snowflake, SQL, MongoDB, Elasticsearch, PySpark, Hadoop, Pinecone

Gen & Agentic AI Libraries: Hugging Face Transformers, Unsloth, CrewAI, Llama.cpp, LangChain, LangGraph, Claude Code, Codex Cli, Hermes, OpenClaw, OpenCode

Soft Skills: Analytical Thinking, Rigor, Project Leadership, Communication, Collaboration

Education

l’X Ecole Polytechnique, Paris-Palaiseau, France *M2 Data Science*

Sept 2024 – Sept 2025

- **Relevant Coursework:** Computer Vision, Optimization for data science, Generative models, Graphs Machine Learning, Causal inference, Big Data, Cloud architectures...

EMINES, Mohammed VI Polytechnic University, Morocco *Master of Engineering in*

Sept 2021 – Sept 2024

Industrial Management, Data Science Minor

- **Relevant Coursework:** Probability Theory and Statistics, Intensive Data-science courses, Macroeconomics and Finance

Lycée CPGE Ibn Timiya, Marrakech, Morocco

Sept 2019 – June 2021

Classes Préparatoires aux Grandes Écoles (CPGE)

Experience

Hager Group – *Quality Control: Anomaly Detection*

Obernai, France

Libraries and skills: PyTorch Lightning, Azure DevOps, Azure IoT Edge, Anomalib

Feb. 2026 – Present

- Developed and productionized computer-vision systems for automated manufacturing conformity inspection, covering dataset preparation, model development, evaluation, deployment, and production support.
- **Relay Vision:** Trained and optimized an EfficientNet-based defect detection model for 2 mm components and resource-constrained IoT Edge deployment, achieving < 1% false rejection rate and 25 PPM undetected defects.
- **Component Control:** Developed an unsupervised anomaly-detection solution using **Dinomaly** anomaly localization through heatmaps, coupled with a **GLM-OCR** module for automated reference-number recognition on inline components.
- Designed and maintained MLOps/DevOps CI/CD pipelines using Azure DevOps, automating the build and deployment of the complete application stack, with model training and lifecycle management on Azure ML.
- Acted as technical owner across both projects, driving technical decisions, coordinating with production teams, translating operator expertise into ML requirements, and contributing to application-level architecture and refactoring.

IDEMIA PUBLIC SECURITY - *Iris Recognition*

Courbevoie, France

Libraries and Skills: PyTorch, PyTorch Lightning, MLflow, Optuna, Accelerate, Autonomy

April 2025 – Oct 2025

- Trained and optimized a **Vision Transformer** for iris identification, achieving a **0.26% FRR** and reducing comparison workload from **16 pairwise checks** to a **single** comparison, dramatically accelerating biometric matching.

CLEVERLYTICS - *Iris Recognition*

Ben Guerir, Morocco

Libraries and Skills: PyTorch, OpenCV, C++, OpenIRIS, Project Leadership

April 2024 – Oct 2024

- Led the end-to-end Iris recognition project and served as main client interface, delivering a **World-Coin** based pipeline that improved **accuracy by 60%** on the **Ubiris** dataset and presenting results to guide biometric system decisions.
- Designed a two-stage DL approach (Autoencoder + Siamese CNN) that cut UBIRIS FRR from **50%** to **20%**, supported by weekly client presentations and clear technical documentation highlighting business impact.

AI/ML Projects

Local Multimodal Agentic RAG System

[GitHub](#) [🔗](#)

Sept 2025

Libraries and Skills: llama.cpp, Qwen, Voxtral, Milvus, Docker, Streamlit, CrewAI

- Built a fully local multimodal RAG system with llama.cpp (Qwen, Voxtral) for document and audio processing, using Milvus vector database for semantic search, deployed via Streamlit and Docker with no external API calls.

Fast style transfer with instance normalization

[GitHub](#) [🔗](#)

May 2025

Libraries and Skills: PyTorch, Git, Docker, Streamlit

- Developed of a real-time style transfer system using a CNN autoencoder, deployed via an interactive Streamlit interface and containerized with Docker for simplified multi-platform deployment.

Fine-tuning a Small Language Model for Summarization [GitHub](#) [🔗](#) [Hugging Face](#) [🔗](#)

April 2025

Libraries and Skills: Unsloth, HG Transformers, PEFT, TRL, llama.cpp, ollama

- Fine-tuned Qwen 2.5-0.5B-Instruct on CNN/DailyMail using LoRA and 4-bit quantization for news summarization (+11-28% ROUGE), with GGUF conversion for llama.cpp/Ollama deployment.